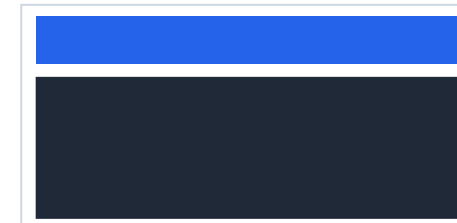


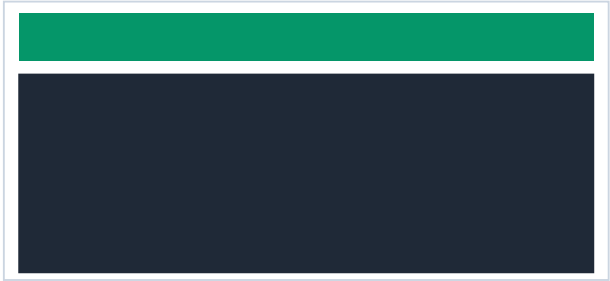
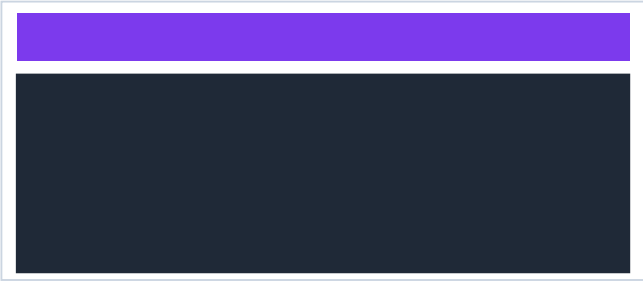
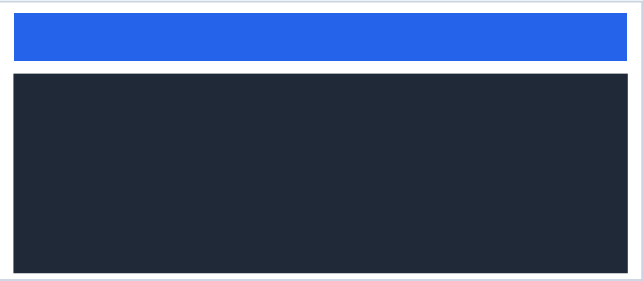


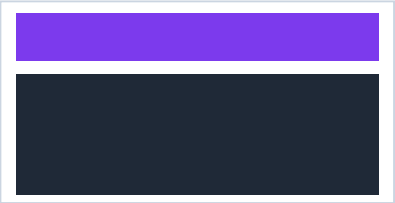
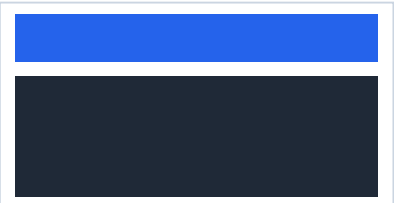
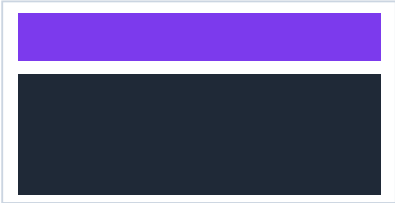
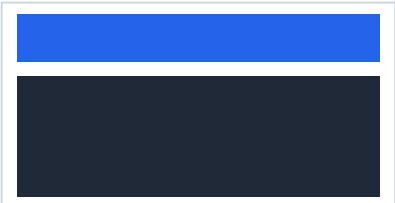
阶段性结束与问题诊断





关键判断：reflection consistency 是反射结构的内部约束，不等价于 full RGB 或 mesh quality。





注意：这里约束的是“可见、几何一致、置信度高的相邻视角对应区域”的反射预测稳定性，

不是假设真实 specular radiance 跨视角严格相等。





		Base	RC	Base	RC	Base	RC	Base	RC
synthetic									
synthetic									



观察：RC ↓ 在两个 dataset 的 train/test Avg. 均下降；PSNR/SSIM/LPIPS 的变化幅度和方向更依赖数据集与 split。

因此主表应作为 per-dataset / per-scene 数值对照，而不是“全面优于 baseline”的证据。



Refl. PSNR : 16 win / 12 loss

Refl. SSIM : 14 win / 14 loss

Refl. LPIPS : 10 win / 18 loss

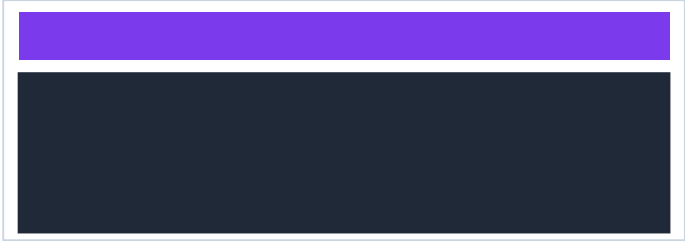
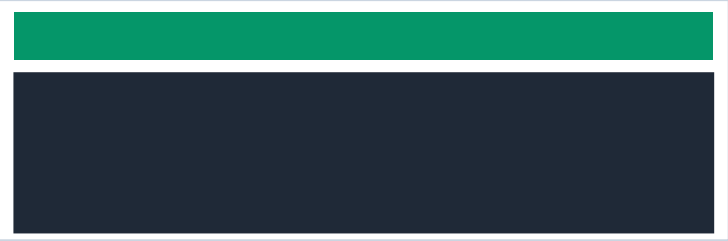
正值表示 RC 更好

结论：反射区域 RGB 指标并未随 consistency 稳定提升。

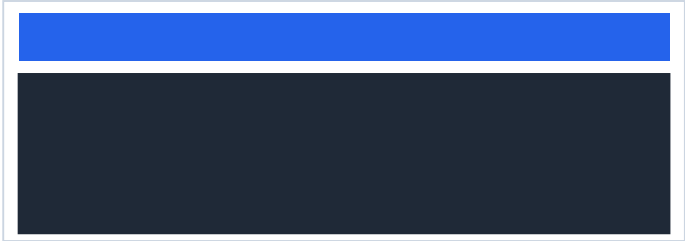
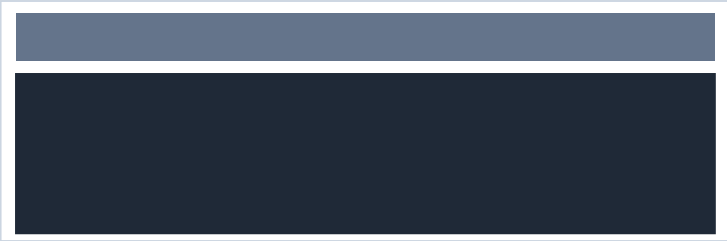
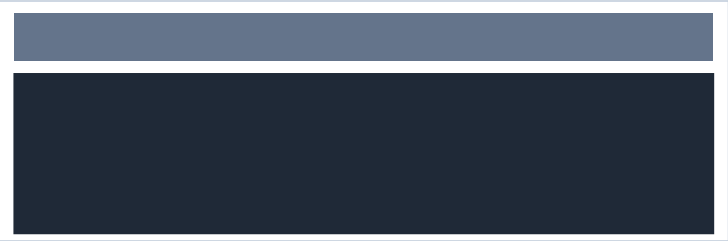
Pearson r : Refl. PSNR 0.01, Refl. SSIM 0.04, Refl. LPIPS 0.10 ; Spearman 同样接近 0。

存在多行 “consistency 改善，但 reflective image quality 下降” 的 trade-off。

- wo_ref 牺牲 consistency，但若干 Repl. metrics 可持平或更好
- wo_conf 削弱 consistency，说明 confidence 有贡献但非充分条件
- rough_only 不能复现 RC 的 reflection consistency 行为
- 消融支持模块边界，不支持“所有指标全面提升”



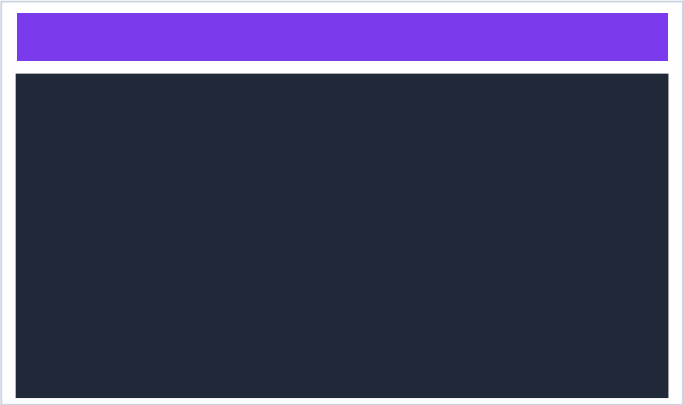
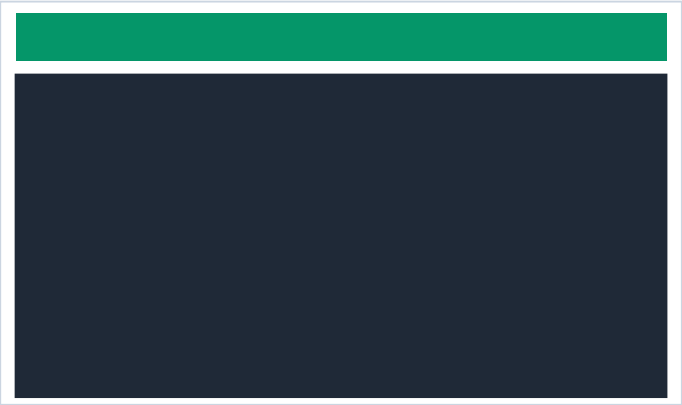
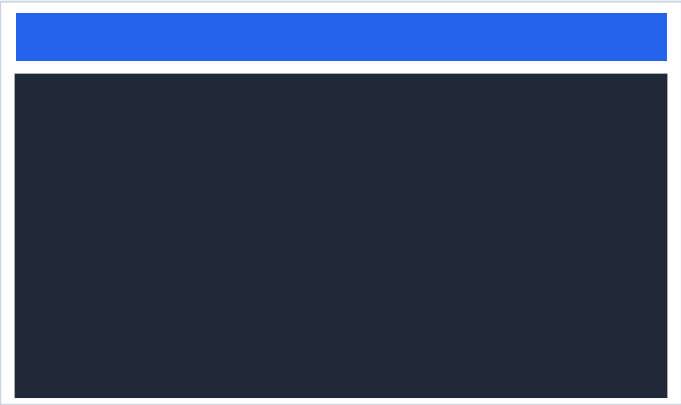
final-RGB 指标



仅提升 reflection consistency 是不够的。

- 当前结果包没有 Chamfer / F-score / normal MAE 等独立几何指标
- confidence-aware geometry filtering 只能作为工程增强模块讨论
- reflection consistency 更像内部诊断指标，而不是最终表面质量指标
- 下一步必须证明 RC inconsistency 与 mesh artifact 有空间对应关系

Unsafe claim：RC 必然提升 mesh quality。Safe claim：mesh quality 需要独立评估。



- Chamfer / F-score / normal MAE
- RC mask 与 mesh artifact overlap
- valid correspondence ratio
- depth pass rate / normal agreement
- pair angle distribution

- mean RC weight / weight sum
- specular confidence distribution
- reflective mask area ratio
- specular variance / highlight sharpness
- lambda_RC / schedule sweep

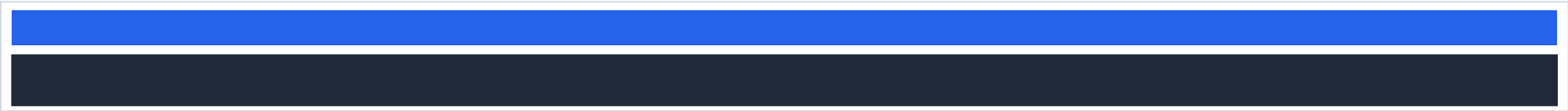
这些诊断可以回答：RC 改善的是反射稳定性，还是确实减少了几何伪影？



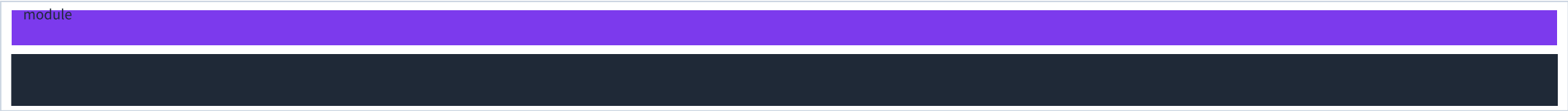
reflection consistency



和 mesh quality 的稳定提升



regularizer / diagnostic



module

aware RC filtering

